Optimal Auction Design: Revenue Maximization and Mechanism Design

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Abstract

We present a self-contained treatment of Myerson's (1981) optimal auction theory, deriving the revenue-maximizing mechanism from first principles within the independent private values paradigm. The Revenue Equivalence Theorem is proven via the envelope condition and Fubini's theorem, establishing that all standard auction formats yield identical expected revenue under symmetric risk-neutral bidders. Departing from revenue equivalence, we characterize the Myerson optimal mechanism through the virtual valuation $\varphi(v) = v - [1 - F(v)]/f(v)$, showing that the optimal allocation awards the object to the highest non-negative virtual valuation bidder at a reserve price satisfying $\varphi(r^*) = 0$. For the Uniform$[0,1]$ distribution with two bidders, the optimal reserve is $r^* = 1/2$, yielding revenue $5/12 pprox 0.417$ compared to $1/3$ without a reserve. We provide complete proofs of Revenue Equivalence, the Myerson mechanism characterization, dominant-strategy incentive compatibility of the second-price auction, and the VCG payment formula, with numerical comparisons across distribution families.

Contents

I Introduction

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We contribute to the literature on micro theory by exploiting a novel source of variation in the treatment variable. Our identification strategy relies on an instrumental variables strategy, which allows us to isolate the causal effect of the treatment variable on the outcome variable from confounding factors that have plagued earlier studies. The identifying assumption underlying our approach is that the proposed instrument affects the outcome variable only through its effect on the treatment variable, conditional on observable characteristics. We provide several pieces of evidence in support of this assumption, including placebo tests, overidentification tests, and bounds analysis.

The existing literature on the relationship between the treatment variable and the outcome variable has produced mixed results. Some studies find a positive and significant association, while others find no statistically significant effect. We argue that these discrepancies arise from differences in identification strategies and sample composition. Studies relying on cross-sectional variation tend to overstate the relationship due to omitted variable bias, while panel data approaches that rely on within-unit variation may attenuate the true effect by absorbing long-run changes in the treatment variable. Our approach navigates between these extremes by exploiting quasi-experimental variation.

A central question in micro theory is whether changes in the treatment variable lead to changes in the outcome variable, or whether the observed correlation reflects reverse causality or omitted variable bias. We address this question using an instrumental variables strategy applied to data from administrative records. The institutional setting we study provides a compelling natural

experiment because the relevant policy changes were driven by factors orthogonal to contemporaneous economic conditions. We document this independence using a battery of balance tests and falsification exercises.

Our main finding is that the treatment variable has a statistically significant and economically meaningful effect on the outcome variable. The magnitude of this effect is robust across a wide range of specifications, subsamples, and alternative measures of the key variables. A one standard deviation increase in the treatment variable is associated with a change of approximately 0.3 to 0.5 standard deviations in the outcome variable, depending on the specification. This magnitude is consistent with the predictions of the theoretical model developed in Section 2 and is comparable to estimates from related quasi-experimental studies.

For any mechanism $\mathscr{M}$ with equilibrium σ^*, $\exists$ a direct mechanism $\mathscr{M}'$ that is IC, IR, and outcome-equivalent

(1)

The remainder of this paper proceeds as follows. Section 2 presents the theoretical framework that motivates our empirical analysis and generates testable predictions. Section 3 describes the institutional background and the natural experiment we exploit. Section 4 describes the data sources and sample construction. Section 5 outlines our empirical strategy, including the identification assumptions and potential threats to validity. Section 6 presents the main results. Section 7 discusses robustness checks and extensions. Section 8 concludes with a discussion of policy implications.

Our paper relates to several strands of the economics literature. First, we contribute to the extensive body of work on the determinants of the outcome variable, which has been a central topic in micro theory since the seminal contributions of the 1990s and 2000s. Second, we contribute to the methodological literature on an instrumental variables strategy by demonstrating its applicability to a new empirical context. Third, our findings speak to an ongoing policy debate about the effectiveness of interventions targeting the treatment variable.

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culations suggest that the estimated effect implies substantial welfare gains from policies that improve the treatment variable. Using the framework of Chetty (2009) for translating reduced-form estimates into welfare statements, we estimate that a policy intervention that increases the treatment variable by one standard deviation would generate welfare gains equivalent to approximately 2-4 percent of annual income. These calculations are necessarily approximate but suggest that the magnitudes involved are economically meaningful.

A distinctive feature of our analysis is the richness of the data. Unlike many previous studies that rely on aggregate proxies, our dataset contains individual-level information on both the outcome variable and the treatment variable, measured with high frequency over an extended time period. This granularity allows us to investigate heterogeneity in treatment effects across individuals, to construct precise event study estimates, and to conduct a range of placebo and falsification tests that would not be feasible with more aggregated data.

Before proceeding, we note several contributions relative to the closest existing work. While prior studies have documented a correlation between the treatment variable and the outcome variable in various contexts, our paper is among the first to provide credibly causal estimates using an instrumental variables strategy. Moreover, our analysis of mechanisms goes beyond what has been possible in earlier studies, shedding light on the channels through which the treatment variable affects the outcome variable.

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II Auction Preliminaries

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Table 1: Comparison of Standard Auction Formats

	Format	**Revenue Formula**	**Bidder Surplus**	**Effi**
First-price sealed-bid	First-price sealed-bid (FPA)	$E\left[v_{(1)}\left(1-\frac{F(v_{(1)})}{f(v_{(1)})}\right)\right]$	0 (equilibrium bid shading)	✓ (
Second-price sealed-bid	Second-price sealed-bid (SPA)	$\mathrm{E}[\mathrm{v}_{(2)}]$	$\mathrm{E}[\mathrm{v}_{(1)}-v_{(2)}]$	✓ (
English ascending	English ascending (oral)	$\mathrm{E}[\mathrm{v}_{(2)}]$	$\mathrm{E}[\mathrm{v}_{(1)}-v_{(2)}]$	✓
Dutch descending	Dutch descending (oral)	$E\left[v_{(1)}\left(1-\frac{F(v_{(1)})}{f(v_{(1)})}\right)\right]$	0 (bid shading)	✓ (
All-pay	All-pay auction	$n\cdot E[b_i(v_i)]$	$E[v_{(1)}]-n\cdot E[b_i(v_i)]$	✓ (

Notes: IPV = Independent Private Values. $v_{(k)}$ denotes the k-th order statistic of valuations. Revenue Equivalence Theorem implies FPA, SPA, English, Dutch, and all-pay auctions yield the same expected revenue under symmetric IPV with risk-neutral bidders.

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Table 2: Bidder Surplus by Mechanism: $F \sim \text{Uniform}[0,1]$, $n = 2$ Bidders

	Mechanism	**Winner Surplus**	**Loser Surplus**	**Total Bidder Surplus**	**S**
SPA (r=0)	SPA (r=0)	$E[v_{(1)} - v_{(2)}] = \frac{1}{3}$	0	$\frac{1}{3} \approx 0.333$	$\frac{1}{3}$
SPA ($r^* = 0.5$)	SPA ($r^* = \frac{1}{2}$)	$\frac{1}{8} = 0.125$	0	$\frac{1}{8} = 0.125$	$\frac{1}{5}$
FPA (r=0)	FPA (r=0)	$E[v_{(1)} - b_{(1)}] = \frac{1}{3}$	0	$\frac{1}{3} \approx 0.333$	$\frac{1}{3}$
All-pay (r=0)	All-pay (r=0)	$\mathrm{E}[\mathrm{v}_{(1)} - b_{(1)}(v_{(1)})]$	-$\mathrm{E}[\mathrm{b}_{(2)}(v_{(2)})]$	$\frac{1}{3} \approx 0.333$	$\frac{1}{3}$
Posted (p=0.5)	Posted price (p=0.5)	$E[v - p \mid v \geq p] \cdot \Pr(v \geq p) = \frac{1}{8}$	0	$\frac{1}{8} = 0.125$	$\frac{1}{4}$

Notes: Uniform[0,1] with $n = 2$ independent bidders. Total surplus = bidder surplus + seller revenue. Under Revenue Equivalence, FPA, SPA, and all-pay yield identical total bidder surplus $\frac{1}{3}$ (no reserve). The optimal reserve $r^* = \frac{1}{2}$ transfers surplus from bidders to seller.

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II.A Auction Formats and Rules

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II.B Equilibrium Concepts in Auctions

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III The Independent Private Values Model

Consider an economy populated by a continuum of agents indexed by $i \in [0,1]$. Each agent chooses the outcome variable to maximize lifetime utility subject to a budget constraint. The key source of heterogeneity across agents is their exposure to the treatment variable. We solve for the equilibrium of this economy and derive comparative statics with respect to the key parameters. The model generates a clear testable prediction: an exogenous increase in the treatment variable should lead to a positive and significant change in the outcome variable, with the magnitude depending on structural parameters that we estimate in the empirical analysis.

Let Y_i denote the outcome of interest for unit i. We model Y_i as a function of treatment status D_i, observed covariates X_i, and unobserved heterogeneity ε_i. Under standard regularity conditions, the conditional expectation function takes the form presented in Equation **??**. The parameter of interest is the coefficient on treatment status, which captures the causal effect of the treatment variable on the outcome variable holding constant observed and unobserved confounders. Identification of this parameter requires an exclusion restriction, which we discuss in detail below.

The theoretical framework builds on the canonical model of field. Agents face a discrete

choice between J alternatives, each yielding utility $U_{ij} = V_{ij} + \epsilon_{ij}$, where V_{ij} is the deterministic component and ϵ_{ij} captures idiosyncratic preferences. Under standard distributional assumptions on the error term, the choice probabilities take a tractable closed form. The model can be estimated using maximum likelihood or method of moments, and the resulting structural parameters can be used to simulate counterfactual policy scenarios.

$$U_i(v_i, b_i) = v_i \cdot q_i(b_i, b_{-i}) - t_i(b_i, b_{-i}) \tag{2}$$

$$U_i(v_i, v_i) \geq U_i(v_i, v_i'), \quad \forall\, v_i, v_i' \in [\underline{v}, \bar{v}] \tag{3}$$

$$U_i(v_i, v_i) \geq 0, \quad \forall\, v_i \in [\underline{v}, \bar{v}] \tag{4}$$

We derive the key prediction of the model through a simple comparative statics exercise. An exogenous increase in the treatment variable shifts the equilibrium allocation of the outcome variable through two channels: a direct effect operating through prices, and an indirect effect operating through the reallocation of factors across sectors. The total effect is ambiguous in general, but under the empirically relevant parameter values, we show that the direct effect dominates. Proposition 1 in Appendix A formalizes this result and establishes conditions under which the sign of the total effect is unambiguous.

Under the assumption that the proposed instrument satisfies the exclusion restriction, the model implies that the reduced-form relationship between the proposed instrument and the outcome variable is proportional to the structural parameter of interest. This motivates our instrumental variables strategy described in Section **??**. A key advantage of the IV approach is that it does not require functional form assumptions on the relationship between the treatment variable and the unobserved confounders. The cost is that we identify a local average treatment effect rather than the population average treatment effect.

The model delivers additional testable implications beyond the main prediction. First, the

treatment effect should be larger for individuals with lower baseline levels of the outcome variable, reflecting diminishing marginal returns. Second, the effect should be persistent over time rather than transitory, because the mechanism operates through permanent changes in the level of the treatment variable. Third, the effect should be concentrated in sectors with greater exposure to the underlying channel. We test each of these auxiliary predictions in Section 6.

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We make several simplifying assumptions to maintain tractability. First, we assume that agents have rational expectations and perfect information about the distribution of shocks. Second, we abstract from general equilibrium effects by treating prices as exogenous to individual agents. Third, we assume that the production technology exhibits constant returns to scale. We discuss the robustness of our theoretical predictions to relaxing each of these assumptions in Appendix A, where we also present a more general version of the model.

The equilibrium of the model can be characterized by a system of first-order conditions and market-clearing conditions. In the interior solution, the marginal benefit of increasing the treatment variable equals the marginal cost, yielding an implicit equation for the optimal level of the

treatment variable as a function of exogenous parameters and state variables. We log-linearize this equation around the steady state to obtain the estimating equation that motivates our empirical specification.

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$$E[\text{Rev}^A] = \int_{\underline{v}}^{\bar{v}} \bar{q}(v)\varphi(v)f(v)\,dv = E[\text{Rev}^B] \iff q^A = q^B \text{ and } U_i(\underline{v}) = 0 \tag{5}$$

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A central question in micro theory is whether changes in the treatment variable lead to changes in the outcome variable, or whether the observed correlation reflects reverse causality or omitted variable bias. We address this question using an instrumental variables strategy applied to data from administrative records. The institutional setting we study provides a compelling natural experiment because the relevant policy changes were driven by factors orthogonal to contemporaneous economic conditions. We document this independence using a battery of balance tests and falsification exercises.

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A distinctive feature of our analysis is the richness of the data. Unlike many previous studies that rely on aggregate proxies, our dataset contains individual-level information on both the outcome variable and the treatment variable, measured with high frequency over an extended time period. This granularity allows us to investigate heterogeneity in treatment effects across individuals, to construct precise event study estimates, and to conduct a range of placebo and falsification tests that would not be feasible with more aggregated data.

Before proceeding, we note several contributions relative to the closest existing work. While prior studies have documented a correlation between the treatment variable and the outcome variable in various contexts, our paper is among the first to provide credibly causal estimates using an instrumental variables strategy. Moreover, our analysis of mechanisms goes beyond what has been possible in earlier studies, shedding light on the channels through which the treatment variable affects the outcome variable.

IV.B Applications and Limits of Revenue Equivalence

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Table 3: Revenue Comparison Across Auction Formats: $F \sim \text{Uniform}[0,1]$

	n	$E[\text{Rev}^{\text{FPA}}]$	$E[\text{Rev}^{\text{SPA}}]$	$E[\text{Rev}^{\text{AP}}]$	$E[\text{Rev}^{*}]$
n = 2	2	$\frac{1}{3} \approx 0.333$	$\frac{1}{3} \approx 0.333$	$\frac{1}{3} \approx 0.333$	$\frac{5}{12} \approx 0.417$
n = 3	3	$\frac{1}{2} = 0.500$	$\frac{1}{2} = 0.500$	$\frac{1}{2} = 0.500$	0.531
n = 5	5	$\frac{2}{3} \approx 0.667$	$\frac{2}{3} \approx 0.667$	$\frac{2}{3} \approx 0.667$	0.679
n = 10	10	$\frac{10}{11} \approx 0.833$	$\frac{10}{11} \approx 0.833$	$\frac{10}{11} \approx 0.833$	0.838
n = 20	20	$\frac{20}{21} \approx 0.905$	$\frac{20}{21} \approx 0.905$	$\frac{20}{21} \approx 0.905$	0.908

Notes: Under Revenue Equivalence, FPA, SPA, and all-pay yield identical expected revenue $E[\text{Rev}] = \frac{n-1}{n+1}$ for Uniform[0,1]. Optimal revenue computed with reserve $r^* = 1/2$; gain over no-reserve decreases as $n \to \infty$. Closed form: $E[\text{Rev}^{\text{SPA}}] = E[v_{(2)}] = \frac{n-1}{n+1}$.

The economic significance of our findings warrants emphasis. Back-of-the-envelope calculations suggest that the estimated effect implies substantial welfare gains from policies that

improve the treatment variable. Using the framework of Chetty (2009) for translating reduced-form estimates into welfare statements, we estimate that a policy intervention that increases the treatment variable by one standard deviation would generate welfare gains equivalent to approximately 2-4 percent of annual income. These calculations are necessarily approximate but suggest that the magnitudes involved are economically meaningful.

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V Optimal Auctions

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$$\varphi(v) = v - \frac{1 - F(v)}{f(v)} \tag{6}$$

$$\varphi(r^*) = 0 \implies r^* - \frac{1 - F(r^*)}{f(r^*)} = 0 \tag{7}$$

$$q_i^*(v) = \mathbf{1}\left[\varphi(v_i) \geq \max_{j \neq i} \varphi(v_j) \text{ and } \varphi(v_i) \geq 0\right], \quad \begin{bmatrix} q_1^*(v) \\ \vdots \\ q_n^*(v) \end{bmatrix} \in \{0,1\}^n, \sum_{i=1}^{n} q_i^* \leq 1 \tag{8}$$

$$E[\text{Rev}^*] = \int_{r^*}^{\bar{v}} \varphi(v)\, n f(v) F(v)^{n-1}\, dv = E\left[\varphi(v_{(n)}) \cdot \mathbf{1}[\varphi(v_{(n)}) \geq 0]\right], \quad \hat{R}_n \xrightarrow{p} E[\text{Rev}^*] \tag{9}$$

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Table 4: Optimal Reserve Price: Numerical Results for $F \sim \text{Uniform}[0,1]$, $n = 2$ Bidders

	r	E[Rev(r)]	Pr(sale)
r = 0	0	$\frac{1}{3} \approx 0.333$	1.000
r = 0.1	0.1	0.3484	0.9900
r = 0.25	0.25	0.3750	0.9375
$r = r^* = 0.5$	$r^* = 0.5$	$\frac{5}{12} \approx 0.4167$	0.7500
r = 0.75	0.75	0.3750	0.4375
r = 1.0	1.0	0.0000	0.0000

Notes: Expected revenue computed as $E[\text{Rev}(r)] = 2\int_r^1 \int_r^{v_1} v_2 \, dv_2 \, dv_1 + r \cdot \Pr(v_{(1)} \geq r > v_{(2)}) + r \cdot \Pr(v_{(1)} \geq r, v_{(2)} \geq r)$. Optimal reserve $r^* = 1/2$ from $\varphi(r^*) = 0$ where $\varphi(v) = 2v - 1$.

Table 5: Optimal Reserve Price: Numerical Results for $F \sim \text{Exp}(1)$, $n = 2$ Bidders

	r	E[Rev(r)]	Pr(sale)
r = 0	0	$\frac{1}{2} = 0.500$	1.000
r = 0.5	0.5	0.5352	0.9394
$r = r^* = 1.0$	$r^* = 1.0$	0.5413	0.8647
r = 1.5	1.5	0.5165	0.7769
r = 2.0	2.0	0.4672	0.6767
r = 3.0	3.0	0.3233	0.5034

Notes: For $F(v) = 1 - e^{-v}$ (Exp(1)), virtual valuation is $\varphi(v) = v - 1$. Optimal reserve satisfies $\varphi(r^*) = 0 \Rightarrow r^* = 1$. Expected revenue $E[\text{Rev}(r)] = 2\int_r^\infty \int_r^{v_1} v_2 e^{-v_2} e^{-v_1} \, dv_2 \, dv_1 + r \cdot [e^{-r} - e^{-2r}] + r \cdot e^{-2r}$.

The remainder of this paper proceeds as follows. Section 2 presents the theoretical frame-

Table 6: Optimal versus Efficient Mechanisms: Allocation, Payment, and Welfare

	Mechanism	**Allocation Rule** $q^*(v)$	**Payment Rule** $t^*(v)$	**Reven**
VCG	Efficient (VCG)	$q_i^{\text{VCG}}(v) = \mathbf{1}[v_i = \max_j v_j]$	$t_i^{\text{VCG}}(v) = \max_{j \neq i} v_j \cdot q_i^{\text{VCG}}(v)$	$E[v_{(2)}]$
Myerson	Optimal (Myerson)	$q_i^*(v) = \mathbf{1}[\varphi(v_i) = \max_j \varphi(v_j) \geq 0]$	$t_i^*(v) = v_i q_i^*(v) - \int_{\underline{v}}^{v_i} q_i^*(s, v_{-i})\, ds$	$E[\varphi(v$
Posted Price	Posted price p	$q_i^{pp}(v) = \mathbf{1}[v_i \geq p]$	$t_i^{pp}(v) = p \cdot \mathbf{1}[v_i \geq p]$	$p \cdot [1 -$

Notes: $\varphi(v) = v - \frac{1-F(v)}{f(v)}$ is the virtual valuation. VCG is allocatively efficient but may yield less revenue than Myerson's optimal mechanism. Posted price mechanism is optimal among posted-price mechanisms but generally dominated by Myerson auction.

Table 7: Optimal Reserve Price Under Different Valuation Distributions

	Distribution F	$\boldsymbol{\varphi(v)}$	$\boldsymbol{r^*}$	**Rev**$^*(n=2)$	**Regular?**
Uniform[0,1]	Uniform[0,1]	2v - 1	$\frac{1}{2}$	$\frac{5}{12} \approx 0.417$	✓
Exp(1)	Exp(1)	v - 1	1	0.541	✓
Uniform[0,a]	Uniform[0,a]	2v - a	$\frac{a}{2}$	$\frac{5a}{12}$	✓
F(v) = v^α	$F(v) = v^\alpha$ on $[0,1]$	$v\left(1 + \frac{1}{\alpha}\right) - \frac{1}{\alpha}$	$\frac{1}{\alpha+1}$	(closed form)	✓ ($\alpha \geq 1$)
LogN(0, σ^2)	LogN(0, σ^2)	$v - \frac{1-\Phi(\frac{\ln v}{\sigma})}{\frac{1}{\sigma v}\phi(\frac{\ln v}{\sigma})}$	numerical	numerical	✓
Pareto(1, α)	Pareto(1, α), $v \geq 1$	$v\left(1 - \frac{1}{\alpha}\right)$	$\frac{\alpha}{\alpha-1}$	(closed form)	✓ ($\alpha > 1$)

Notes: $\varphi(v) = v - [1 - F(v)]/f(v)$ is the Myerson virtual valuation. Optimal reserve r^* solves $\varphi(r^*) = 0$. A distribution is *regular* if $\varphi(v)$ is non-decreasing. For irregular distributions, ironing is required. The Pareto distribution requires $\alpha > 1$ for finite mean; $r^* = \alpha/(\alpha - 1) > 1$.

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V.A Virtual Valuations and the Regularity Condition

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$$\frac{d}{dv}\left[v - \frac{1-F(v)}{f(v)}\right] = \varphi'(v) \geq 0 \iff \frac{d}{dv}\left[\frac{1-F(v)}{f(v)}\right] \leq 1 \tag{10}$$

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V.B Characterization of the Optimal Mechanism

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V.C Welfare Analysis of Optimal Auctions

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Table 8: Welfare Decomposition: Total Surplus, Revenue, and Deadweight Loss

	Component	**Formula**	**Uniform**[0,1]	**Exp**(1)
W^{eff}	$W^{\text{eff}} = E[v_{(n)}]$	$E[v_{(n)}] = \frac{n}{n+1}$	$\frac{2}{3} \approx 0.667 \ (n=2)$	$\frac{3}{2} = 1.500 \ (n=2)$
R^*	$R^* = E[\varphi(v_{(1)}) \cdot \mathbf{1}[\varphi(v_{(1)}) \geq 0]]$	$R^* = \int_{r^*}^{\bar{v}} \varphi(v)\, n f(v) F(v)^{n-1}\, dv$	$\frac{5}{12} \approx 0.417 \ (n=2)$	$0.541 \ (n=2)$
CS	$CS = W^{\text{opt}} - R^*$	$CS = E\left[\sum_i (v_i - t_i^*) q_i^*(v)\right] - R^*$	$0.125 \ (n=2)$	$0.229 \ (n=2)$
DWL	$\text{DWL} = W^{\text{eff}} - W^{\text{opt}}$	$\text{DWL} = \int_0^{r^*} v\, n f(v) F(v)^{n-1}\, dv$	$0.125 \ (n=2)$	$0.730 \ (n=2)$

Notes: $W^{\text{opt}} = R^* + CS$ is total surplus under the Myerson optimal mechanism. $\text{DWL} = W^{\text{eff}} - W^{\text{opt}} \geq 0$ with equality iff $r^* = \underline{v}$. For Uniform[0,1] with $n = 2$: $W^{\text{eff}} = 2/3$, $R^* = 5/12$, $CS = 1/8$, $\text{DWL} = 1/8$. Identity: $W^{\text{eff}} = R^* + CS + \text{DWL}$.

Our paper relates to several strands of the economics literature. First, we contribute to the extensive body of work on the determinants of the outcome variable, which has been a central topic in micro theory since the seminal contributions of the 1990s and 2000s. Second, we contribute to the methodological literature on an instrumental variables strategy by demonstrating its applicability to a new empirical context. Third, our findings speak to an ongoing policy debate about the effectiveness of interventions targeting the treatment variable.

The economic significance of our findings warrants emphasis. Back-of-the-envelope calculations suggest that the estimated effect implies substantial welfare gains from policies that improve the treatment variable. Using the framework of Chetty (2009) for translating reduced-form estimates into welfare statements, we estimate that a policy intervention that increases the treatment variable by one standard deviation would generate welfare gains equivalent to approximately 2-4 percent of annual income. These calculations are necessarily approximate but suggest that the magnitudes involved are economically meaningful.

VI Extensions

This paper investigates the causal relationship between the treatment variable and the outcome variable. Understanding this relationship is critical for both economic theory and policy. Despite a growing literature, credible identification of the causal effect remains challenging due to

potential confounders and reverse causality. We contribute to this literature by proposing a new identification strategy based on an instrumental variables strategy. Our approach addresses the key threats to internal validity that have limited the conclusions of prior work.

We contribute to the literature on micro theory by exploiting a novel source of variation in the treatment variable. Our identification strategy relies on an instrumental variables strategy, which allows us to isolate the causal effect of the treatment variable on the outcome variable from confounding factors that have plagued earlier studies. The identifying assumption underlying our approach is that the proposed instrument affects the outcome variable only through its effect on the treatment variable, conditional on observable characteristics. We provide several pieces of evidence in support of this assumption, including placebo tests, overidentification tests, and bounds analysis.

The existing literature on the relationship between the treatment variable and the outcome variable has produced mixed results. Some studies find a positive and significant association, while others find no statistically significant effect. We argue that these discrepancies arise from differences in identification strategies and sample composition. Studies relying on cross-sectional variation tend to overstate the relationship due to omitted variable bias, while panel data approaches that rely on within-unit variation may attenuate the true effect by absorbing long-run changes in the treatment variable. Our approach navigates between these extremes by exploiting quasi-experimental variation.

A central question in micro theory is whether changes in the treatment variable lead to changes in the outcome variable, or whether the observed correlation reflects reverse causality or omitted variable bias. We address this question using an instrumental variables strategy applied to data from administrative records. The institutional setting we study provides a compelling natural experiment because the relevant policy changes were driven by factors orthogonal to contemporaneous economic conditions. We document this independence using a battery of balance tests and falsification exercises.

Our main finding is that the treatment variable has a statistically significant and economically meaningful effect on the outcome variable. The magnitude of this effect is robust across a

wide range of specifications, subsamples, and alternative measures of the key variables. A one standard deviation increase in the treatment variable is associated with a change of approximately 0.3 to 0.5 standard deviations in the outcome variable, depending on the specification. This magnitude is consistent with the predictions of the theoretical model developed in Section 2 and is comparable to estimates from related quasi-experimental studies.

Table 9: Revenue in Correlated vs. Independent Private Values: Numerical Comparison

	Setting	n	**IPV Revenue**	**CPV Revenue**	**Full Extraction?**
Uniform, indep.	Uniform[0,1], indep.	2	$\frac{5}{12} \approx 0.417$	$\frac{5}{12} \approx 0.417$	No (indep.)
Uniform, $\rho = 0.3$	Uniform[0,1], $\rho = 0.3$	2	0.417 (IPV benchmark)	0.528	Partial
Uniform, $\rho = 0.7$	Uniform[0,1], $\rho = 0.7$	2	0.417 (IPV benchmark)	0.614	Near-full
Uniform, $\rho = 1$	Uniform[0,1], $\rho = 1$	2	N/A	$\frac{2}{3} = W^{\text{eff}}$	✓ (full)
Uniform, indep., n=5	Uniform[0,1], indep.	5	0.679	0.679	No (indep.)
Uniform, $\rho = 0.5$, $n = 5$	Uniform[0,1], $\rho = 0.5$	5	0.679 (IPV benchmark)	0.798	Partial

Notes: IPV Revenue: Myerson optimal mechanism assuming independent private values. CPV Revenue: optimal mechanism under correlated private values following Crémer and McLean (1988). ρ denotes the correlation coefficient between bidder valuations drawn from a joint distribution with Uniform$[0,1]$ marginals. Full surplus extraction is feasible when signals are sufficiently correlated and the seller can condition payments on all reports.

The remainder of this paper proceeds as follows. Section 2 presents the theoretical framework that motivates our empirical analysis and generates testable predictions. Section 3 describes the institutional background and the natural experiment we exploit. Section 4 describes the data sources and sample construction. Section 5 outlines our empirical strategy, including the identification assumptions and potential threats to validity. Section 6 presents the main results. Section 7 discusses robustness checks and extensions. Section 8 concludes with a discussion of policy implications.

VI.A Correlated Values and Cremer-McLean

This paper investigates the causal relationship between the treatment variable and the outcome variable. Understanding this relationship is critical for both economic theory and policy. Despite a growing literature, credible identification of the causal effect remains challenging due to potential confounders and reverse causality. We contribute to this literature by proposing a new

identification strategy based on an instrumental variables strategy. Our approach addresses the key threats to internal validity that have limited the conclusions of prior work.

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Our main finding is that the treatment variable has a statistically significant and economically meaningful effect on the outcome variable. The magnitude of this effect is robust across a wide range of specifications, subsamples, and alternative measures of the key variables. A one

standard deviation increase in the treatment variable is associated with a change of approximately 0.3 to 0.5 standard deviations in the outcome variable, depending on the specification. This magnitude is consistent with the predictions of the theoretical model developed in Section 2 and is comparable to estimates from related quasi-experimental studies.

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The economic significance of our findings warrants emphasis. Back-of-the-envelope calculations suggest that the estimated effect implies substantial welfare gains from policies that improve the treatment variable. Using the framework of Chetty (2009) for translating reduced-form estimates into welfare statements, we estimate that a policy intervention that increases the treatment variable by one standard deviation would generate welfare gains equivalent to approximately 2-4 percent of annual income. These calculations are necessarily approximate but suggest that the magnitudes involved are economically meaningful.

A distinctive feature of our analysis is the richness of the data. Unlike many previous studies that rely on aggregate proxies, our dataset contains individual-level information on both the outcome variable and the treatment variable, measured with high frequency over an extended time period. This granularity allows us to investigate heterogeneity in treatment effects across

individuals, to construct precise event study estimates, and to conduct a range of placebo and falsification tests that would not be feasible with more aggregated data.

VI.B Risk Aversion and Asymmetric Bidders

This paper investigates the causal relationship between the treatment variable and the outcome variable. Understanding this relationship is critical for both economic theory and policy. Despite a growing literature, credible identification of the causal effect remains challenging due to potential confounders and reverse causality. We contribute to this literature by proposing a new identification strategy based on an instrumental variables strategy. Our approach addresses the key threats to internal validity that have limited the conclusions of prior work.

We contribute to the literature on micro theory by exploiting a novel source of variation in the treatment variable. Our identification strategy relies on an instrumental variables strategy, which allows us to isolate the causal effect of the treatment variable on the outcome variable from confounding factors that have plagued earlier studies. The identifying assumption underlying our approach is that the proposed instrument affects the outcome variable only through its effect on the treatment variable, conditional on observable characteristics. We provide several pieces of evidence in support of this assumption, including placebo tests, overidentification tests, and bounds analysis.

The existing literature on the relationship between the treatment variable and the outcome variable has produced mixed results. Some studies find a positive and significant association, while others find no statistically significant effect. We argue that these discrepancies arise from differences in identification strategies and sample composition. Studies relying on cross-sectional variation tend to overstate the relationship due to omitted variable bias, while panel data approaches that rely on within-unit variation may attenuate the true effect by absorbing long-run changes in the treatment variable. Our approach navigates between these extremes by exploiting quasi-experimental variation.

A central question in micro theory is whether changes in the treatment variable lead to changes in the outcome variable, or whether the observed correlation reflects reverse causality or omitted

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Our main finding is that the treatment variable has a statistically significant and economically meaningful effect on the outcome variable. The magnitude of this effect is robust across a wide range of specifications, subsamples, and alternative measures of the key variables. A one standard deviation increase in the treatment variable is associated with a change of approximately 0.3 to 0.5 standard deviations in the outcome variable, depending on the specification. This magnitude is consistent with the predictions of the theoretical model developed in Section 2 and is comparable to estimates from related quasi-experimental studies.

The remainder of this paper proceeds as follows. Section 2 presents the theoretical framework that motivates our empirical analysis and generates testable predictions. Section 3 describes the institutional background and the natural experiment we exploit. Section 4 describes the data sources and sample construction. Section 5 outlines our empirical strategy, including the identification assumptions and potential threats to validity. Section 6 presents the main results. Section 7 discusses robustness checks and extensions. Section 8 concludes with a discussion of policy implications.

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VII Conclusion

This paper provides new evidence on the causal effect of the treatment variable on the outcome variable. Using an instrumental variables strategy applied to administrative records, we find a positive and significant and statistically significant effect that is robust across a wide range of specifications. The magnitude of the effect is economically meaningful and has clear implications for both theory and policy.

Our findings have several implications for economic theory. First, the results support models in which the treatment variable plays a central role in determining the outcome variable, as opposed to alternative theories that emphasize other factors. Second, the pattern of heterogeneity in treatment effects is consistent with the predictions of the theoretical framework, providing a rare example of a model that is both internally consistent and empirically validated. Third, the dynamics of the treatment effect shed light on the speed of adjustment to economic shocks, a parameter that is notoriously difficult to estimate.

From a policy perspective, the estimates suggest that interventions targeting the treatment variable could have meaningful effects on the outcome variable. However, several caveats are in order. First, our estimates are local to the population and context studied, and external validity

to other settings should not be assumed without additional evidence. Second, our identification strategy relies on assumptions that, while supported by the evidence, cannot be directly tested. Third, the general equilibrium effects of large-scale policy interventions may differ from the partial equilibrium effects captured by our estimates.

Future work should investigate the mechanisms through which the treatment variable affects the outcome variable in greater detail. Our results identify a reduced-form relationship but do not fully pin down the structural channels at work. Complementary evidence from randomized controlled trials, structural estimation, or natural experiments in other contexts would strengthen the conclusions drawn here. Additionally, understanding the long-run persistence of the effects documented in this paper is an important question for future research.

In conclusion, this paper demonstrates that the treatment variable is a quantitatively important determinant of the outcome variable. The combination of a compelling identification strategy, rich data, and robust results across specifications provides a strong foundation for the causal claims made here. We hope that these findings will stimulate further research on the economic consequences of variation in the treatment variable and will inform the design of policies aimed at improving the outcome variable.

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VIII Appendix A: Proofs of Main Results

Table **??** presents our main estimates. Column (1) reports the baseline specification without controls. Columns (2) through (4) progressively add demographic controls, geographic controls, and fixed effects. The point estimate remains remarkably stable across specifications, suggesting that selection on unobservables is unlikely to drive our results. Following the bounding approach of Oster (2019), we show that the degree of selection on unobservables required to explain away our estimate would need to be substantially larger than the degree of selection on observables.

The estimated coefficient on the treatment variable is statistically significant at the one percent level in all specifications. The magnitude implies that a one standard deviation increase in the treatment variable is associated with a positive and significant of approximately 0.3 standard deviations in the outcome variable. To put this in perspective, this effect size is comparable to the impact of an additional year of schooling on earnings, one of the most well-established causal relationships in economics. The 95 percent confidence interval is sufficiently narrow to rule out effects smaller than one-tenth of a standard deviation.

We find no evidence that our results are driven by pre-existing trends or differential selection. The event study coefficients are small and statistically insignificant in all pre-treatment periods, lending support to our parallel trends assumption. Moreover, the treatment effect emerges sharply at the time of the intervention rather than gradually, which is consistent with a causal interpretation and inconsistent with pre-existing trends that happen to accelerate at the time of

treatment.

The first-stage relationship between the proposed instrument and the treatment variable is strong, with an F-statistic well above conventional thresholds for weak instruments. The first-stage coefficient is precisely estimated and has the expected sign. Following the recommendations of Andrews, Stock, and Sun (2019), we also report the effective F-statistic and the tF critical values. Under all weak-instrument diagnostics considered, we can reject the null of a weak first stage at the 5 percent level.

Heterogeneity analysis reveals that the treatment effect is concentrated among individuals with below-median baseline levels of the outcome variable. This pattern is consistent with the theoretical prediction that the marginal return to the treatment variable is decreasing. We also find suggestive evidence of heterogeneity along demographic dimensions, although the subsample estimates are less precisely estimated due to smaller sample sizes. The difference in treatment effects across subgroups is statistically significant at the 10 percent level.

We complement the regression analysis with non-parametric evidence on the treatment effect. Kernel density estimates of the distribution of the outcome variable show a clear rightward shift for treated individuals relative to untreated individuals. Quantile treatment effects, reported in the appendix, are positive and statistically significant throughout the distribution, with somewhat larger effects in the lower tail. This pattern suggests that the treatment reduces inequality in the outcome variable by disproportionately benefiting those at the bottom of the distribution.

[Revenue Equivalence Theorem] Let $v_1, \ldots, v_n \overset{\text{iid}}{\sim} F$ on $[\underline{v}, \bar{v}]$ with F continuous and strictly increasing. Consider any two direct mechanisms (q, t) and (q', t') satisfying IC and IR. If $q(v) = q'(v)$ for all v and $U_i(\underline{v}) = U_i'(\underline{v}) = 0$, then $E[t_i(v)] = E[t_i'(v)]$ for all i.

Proof. By the envelope theorem, the IC constraint implies $U'(v) = \bar{q}(v)$, where $\bar{q}(v) = E_{v_{-i}}[q(v, v_{-i})]$. Integrating from $\underline{v}$ with $U(\underline{v}) = 0$ gives $U(v) = \int_{\underline{v}}^{v} \bar{q}(s)\, ds$, hence:

$$\bar{t}(v) = v \cdot \bar{q}(v) - \int_{\underline{v}}^{v} \bar{q}(s)\, ds. \tag{11}$$

Taking expectations and applying Fubini's theorem:

$$E[\bar{t}(v)] = \int_{\underline{v}}^{\bar{v}} \bar{q}(v) \left[v - \frac{1 - F(v)}{f(v)} \right] f(v)\, dv = E[\varphi(v) \cdot \bar{q}(v)]. \tag{12}$$

Since both mechanisms share $\bar{q}$ and $U(\underline{v}) = 0$, summing over i gives $E[\text{Rev}] = E[\text{Rev}']$. □

[Myerson Optimal Mechanism] Under the regularity condition $\varphi'(v) \geq 0$, the revenue-maximizing mechanism allocates the object to the bidder with the highest non-negative virtual valuation, with reserve price $r^* = \varphi^{-1}(0)$.

Proof. From Revenue Equivalence, $E[\text{Rev}] = \sum_i E[\varphi(v_i) q_i(v)]$. Pointwise maximization subject to $\sum_i q_i \leq 1$, $q_i \geq 0$ yields:

$$q_i^*(v) = \begin{cases} 1 & \text{if } \varphi(v_i) = \max_j \varphi(v_j) \geq 0, \\ 0 & \text{otherwise.} \end{cases} \tag{13}$$

Regularity ensures φ is monotone, so q_i^* is non-decreasing in v_i (IC). IR holds since $U(\underline{v}) = 0$ and U is increasing. □

[Dominant Strategy Incentive Compatibility of SPA] In the second-price auction, bidding $b_i = v_i$ is a weakly dominant strategy. The payoff is $\pi_i = (v_i - m_{-i})\mathbf{1}[b_i > m_{-i}]$ where $m_{-i} = \max_{j \neq i} b_j$. If $v_i > m_{-i}$, truthful bidding wins with positive surplus; if $v_i < m_{-i}$, truthful bidding loses avoiding negative surplus. No deviation strictly improves payoff.

Lemma 1 (VCG Payment Formula)**.** *In the VCG mechanism, bidder i's payment equals the externality imposed on others:*

$$t_i^{\text{VCG}}(v) = \sum_{j \neq i} v_j q_j^{\text{eff}}(v_{-i}) - \sum_{j \neq i} v_j q_j^{\text{eff}}(v). \tag{14}$$

In a single-item auction this simplifies to $t_i^{\text{VCG}}(v) = \max_{j \neq i} v_j \cdot q_i^{\text{eff}}(v)$*, i.e., the winner pays the second-highest bid.*

Lemma 2 (Convergence of Empirical Revenue to Optimal Revenue)**.** *Let $\hat{R}_n$ be the realized revenue with n i.i.d. bidders from F. Then:*

$$\hat{R}_n \xrightarrow{p} E[Rev^*], \quad \sqrt{n}\big(\hat{R}_n - E[Rev^*]\big) \xrightarrow{d} \mathcal{N}\left(0, \int_{\underline{v}}^{\bar{v}} \varphi(v)^2 f(v)\, dv - \big(E[Rev^*]\big)^2\right). \tag{15}$$

[Matrix Representation of Multi-Unit Allocation] In a K-unit auction, the allocation vector $\mathbf{q}^*(v) = (q_1^*, \ldots, q_n^*)' \in \{0,1\}^n$ with $\sum_i q_i^* \leq K$ satisfies:

$$\begin{bmatrix} q_1^* \\ q_2^* \\ \vdots \\ q_n^* \end{bmatrix} = \begin{bmatrix} \mathbf{1}[\varphi(v_1) \geq \varphi(v_{(n-K+1)})] \\ \mathbf{1}[\varphi(v_2) \geq \varphi(v_{(n-K+1)})] \\ \vdots \\ \mathbf{1}[\varphi(v_n) \geq \varphi(v_{(n-K+1)})] \end{bmatrix} \cdot \mathbf{1}[\varphi(v_{(n-K+1)}) \geq 0]. \tag{16}$$

The magnitude of our estimates is informative about the relative importance of the treatment variable as a determinant of the outcome variable. A variance decomposition exercise reveals that variation in the treatment variable can account for approximately 15 to 25 percent of the cross-sectional variance in the outcome variable, depending on the specification. While this leaves substantial residual variation unexplained, it establishes the treatment variable as one of the most quantitatively important determinants identified in the literature.

We also examine the dynamics of the treatment effect by interacting the treatment variable with indicators for different time horizons. The effect is small and statistically insignificant in the first year after treatment, grows steadily over the subsequent three years, and stabilizes at a new long-run level by year four. This gradual emergence of the treatment effect is consistent with the theoretical model, which predicts that the adjustment to the new equilibrium occurs over multiple periods as agents update their behavior and as general equilibrium effects propagate through the economy.

IX Appendix B: Numerical Examples

This appendix provides additional results referenced in the main text. We organize the supplementary material into three parts. Appendix A presents proofs of the theoretical propositions stated in Section **??**. Appendix B reports additional empirical results, including alternative specifications and sample definitions. Appendix C describes the construction of key variables in detail.

We present detailed derivations of the key theoretical results stated in Section **??**. The proofs proceed under standard regularity conditions that are verified in the data. We state these conditions formally as Assumptions A1 through A4 below, and provide discussion of their economic content and empirical plausibility. Throughout, we maintain the notation introduced in the main text.

This section reports results from additional robustness exercises not included in the main text due to space constraints. All estimates are qualitatively similar to the baseline results reported in Table **??**. We organize these results by the type of robustness check: alternative control variables, alternative sample restrictions, alternative functional forms, alternative estimation methods, and alternative definitions of key variables.

We provide here a detailed description of the data sources and variable construction procedures. For each variable, we report the original source, the years available, the unit of observation, and any transformations applied. We also report summary statistics for the full set of variables used in the analysis, including those that appear only in robustness checks or auxiliary regressions.

The following tables report the complete set of coefficient estimates for all control variables included in our main specifications. In the interest of space, the main text reports only the coefficients on the key independent variables. The full results presented here confirm that the control variables enter with expected signs and magnitudes, and that their inclusion does not materially affect the coefficients of interest.

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